

CLASS 11TH CHEMISTRY | CRUCIAL BOARD QUESTIONS (CYQS) | JAM QUESTION BANK

PHYSICAL & INORGANIC CHEMISTRY - LONG ANSWER QUESTIONS (5 MARKS)

- Q1.** State and explain the laws of chemical combination (Law of conservation of mass, definite proportions, multiple proportions).
- Q2.** Write the main postulates of Dalton's Atomic Theory and its limitations.
- Q3.** Explain the following concentration terms:
(I) Molarity
(II) Mole fraction
(III) Mass percentage of solution
- Q4.** What are the postulates of Bohr's Atomic theory? Explain the postulate of 'Quantization of angular momentum'.
- Q5.** What are Quantum numbers? Explain the principal, azimuthal, magnetic, and spin quantum numbers in detail.
- Q6.** What is photoelectric effect? What are the key observations and conclusions drawn from photoelectric effect studies?
- Q7.** Explain the hydrogen spectrum and its spectral series (Lyman, Balmer, Paschen, Brackett, Pfund). Write the Rydberg formula for the wave number of a spectral line.
- Q8.** Explain the principles of electronic configuration:
(I) Aufbau principle
(II) Pauli's exclusion principle
(III) Hund's rule of maximum multiplicity
- Q9.** Define ionization enthalpy. What are the factors which affect ionization enthalpy? Give its periodic variation:
(I) Across a period
(II) Down a group
- Q10.** Explain atomic radius with definitions and examples:
(I) Covalent radius
(II) Metallic radius
(III) Van der Waals radius
- Q11.** Explain the following periodic terms:
(I) Shielding / Screening effect
(II) Effective nuclear charge (Z_{eff})
(III) Electron gain enthalpy
(IV) Electronegativity
- Q12.** Explain Molecular Orbital Theory (MOT) and draw the molecular orbital energy level diagram for O_2 and N_2 molecules.
- Q13.** What is hybridisation? What are its types (sp , sp^2 , sp^3)? Explain the formation and tetrahedral geometry of methane (CH_4) on the basis of hybridisation.

- Q14.** What is VSEPR theory? Explain the geometry and shapes of the following molecules using VSEPR:
(I) H₂O (Bent / V-shaped)
(II) NH₃ (Trigonal Pyramidal)
(III) SF₆ (Octahedral)
(IV) PCl₅ (Trigonal Bipyramidal)
(V) CO₂ (Linear)
- Q15.** State and explain the First Law of Thermodynamics. Give its different mathematical formulations ($\Delta U = q + w$).
- Q16.** What is enthalpy / heat of reaction? Explain:
(I) Heat of formation ($\Delta_f H$)
(II) Heat of combustion ($\Delta_c H$)
(III) Heat of neutralization
- Q17.** State Hess's Law of Constant Heat Summation. Explain the Born-Haber cycle for NaCl as an application of Hess's Law.
- Q18.** State and explain Le-Chatelier's principle with respect to the effect of:
(I) Pressure
(II) Temperature
(III) Concentration on chemical equilibrium
- Q19.** What is a buffer solution? What are its types? Explain the mechanism of acidic and basic buffer action with examples.
- Q20.** Explain different concepts of acids and bases:
(I) Arrhenius concept
(II) Lewis concept
(III) Bronsted-Lowry protonic concept
- Q21.** Explain different types of redox reactions with balanced chemical equations (Combination, Decomposition, Displacement, and Disproportionation reactions).

PHYSICAL & INORGANIC CHEMISTRY - SHORT ANSWER QUESTIONS (3 MARKS)

- Q22.** Calculation of oxidation number of designated atoms in various compounds (with illustrative examples like KMnO₄, K₂Cr₂O₇, H₂SO₄).
- Q23.** Balancing of redox reactions by oxidation number method in acidic and basic media (Step-by-step examples).
- Q24.** Define oxidizing and reducing agents with suitable redox reaction examples.
- Q25.** State the characteristics of chemical equilibrium and equilibrium constants (K_c and K_p).
- Q26.** What is common ion effect? Discuss its application in qualitative inorganic salt analysis and solubility control.
- Q27.** Calculation of pH and pOH of strong and weak acid/base solutions (Numerical problems).
- Q28.** Calculation of equilibrium constant K_c and equilibrium concentrations from given initial concentrations (Numerical problems).
- Q29.** Calculation of PV-work of expansion and Gibbs free energy change (ΔG) (Numerical problems).

- Q30.** Calculation of Delta H from Delta U using relation: $\Delta H = \Delta U + \Delta n_g \cdot RT$ (Numerical problems).
- Q31.** Define and distinguish between thermodynamic systems (open, closed, isolated) and processes (isothermal, adiabatic, isobaric, isochoric).
- Q32.** What are spontaneous and non-spontaneous processes? Explain with entropy criteria (Second law of thermodynamics).
- Q33.** What is the Gibbs-Helmholtz equation ($\Delta G = \Delta H - T \cdot \Delta S$)? Explain the effect of temperature on the spontaneity of a chemical reaction.
- Q34.** What is an electrovalent / ionic bond? State the favorable energetic conditions for ionic bond formation.
- Q35.** Differentiate between sigma (σ) bond and pi (π) bond with orbital overlapping sketches.
- Q36.** Why is H₂O a liquid with high boiling point while H₂S is a gas at room temperature? Explain on the basis of hydrogen bonding.
- Q37.** Draw the Lewis dot structures of F₂, H₂O, NH₃, and SO₃ molecules.
- Q38.** What are covalent bond parameters (bond length, bond angle, bond enthalpy, bond order)? Explain briefly.
- Q39.** Define dipole moment of a molecule. Write its mathematical formula and explain why CO₂ and BF₃ have zero dipole moment.
- Q40.** What are s-block and p-block elements? Give their general electronic configurations and characteristic physical/chemical properties.
- Q41.** What are Lanthanoids and Actinoids? State their position in the periodic table and general properties.
- Q42.** Explain the following anomalies using electronic configurations:
(I) Boron has lower ionization enthalpy than Beryllium
(II) Oxygen has lower ionization enthalpy than Nitrogen
(III) Fluorine has lower electron gain enthalpy than Chlorine
- Q43.** Numerical problems based on the Mole Concept, molar mass, and Avogadro's number calculation.
- Q44.** Determination of empirical and molecular formula from elemental percentage composition data (Numerical problems).
- Q45.** Define limiting reagent and excess reagent. Explain how to determine the limiting reagent in stoichiometric calculations (with numerical example).

ORGANIC CHEMISTRY - PRINCIPLES, MECHANISMS & HYDROCARBONS

- Q46.** Determine the hybridisation state of each carbon atom in the following compounds:
(a) CH₂=C=O (Ketene)
(b) CH₃-CH=CH₂ (Propene)
(c) (CH₃)₂CO (Acetone)
(d) CH₂=CH-CN (Acrylonitrile)
(e) C₆H₆ (Benzene)
(f) CH₂=C=CH₂ (Allene)

- Q47.** Write the systematic IUPAC names of the following organic compounds:
- Isopropylbenzene (Cumene)
 - 3-bromopropanenitrile
 - 4-methylpent-2-ene
 - 1-chloro-3-bromobenzene
 - 4-chlorobutanal
 - 2,2-dichloroethanol
- Q48.** Briefly describe electron displacement effects in covalent bonds:
- Inductive Effect (+I and -I effects with examples)
 - Electromeric Effect (+E and -E effects with examples)
- Q49.** Write a detailed note on Hyperconjugation effect (Baker-Nathan effect / no-bond resonance) and its role in alkene stability.
- Q50.** Give the correct order of stability among conjugated alkenes and substituted carbocations with theoretical justification.
- Q51.** Explain the terms Aromatic, Anti-aromatic, and Non-aromatic compounds using Huckel's ($4n + 2$) pi electron rule with examples.
- Q52.** Explain the orbital structure, geometry, and relative stability order of carbocations, carbanions, and carbon free radicals.
- Q53.** Arrange the following carbocations in order of decreasing stability with explanation based on hyperconjugation and inductive effect:
- Ethyl carbocation: $\text{CH}_3\text{-CH}_2^+$
 - Isopropyl carbocation: $(\text{CH}_3)_2\text{CH}^+$
 - Tert-butyl carbocation: $(\text{CH}_3)_3\text{C}^+$
 - n-propyl carbocation: $\text{CH}_3\text{-CH}_2\text{-CH}_2^+$
- Q54.** Explain the mechanism of addition reactions of alkenes:
- Addition of dihydrogen (Catalytic hydrogenation)
 - Addition of halogen (Bromination test)
 - Addition of hydrogen halide (HX)
 - Addition of HBr to unsymmetrical alkene: Markownikoff's rule vs Peroxide / Kharasch effect
 - Addition of concentrated H_2SO_4
 - Acid-catalyzed hydration of alkenes to alcohols
- Q55.** Write a short note on the ozonolysis of alkenes. How is ozonolysis used to locate the position of double bonds in alkenes?
- Q56.** Describe electrophilic aromatic substitution reactions of benzene with detailed reaction mechanism (Generation of electrophile, carbocation intermediate formation, proton loss):
- Halogenation
 - Nitration
 - Sulphonation
 - Friedel-Crafts Alkylation & Acylation
- Q57.** Explain the conformational isomerism in Ethane. Draw Sawhorse and Newman projections for Eclipsed and Staggered conformations, and compare their relative stabilities.
- Q58.** How will you carry out the following conversions starting from Benzene:
- Benzene into p-nitrobromobenzene
 - Benzene into m-nitrochlorobenzene
 - Benzene into p-nitrotoluene
 - Benzene into Acetophenone

- Q59.** How would you synthesize Benzene from the following starting materials:
- (I) Ethyne (Cyclic polymerization by passing through red-hot iron tube at 873 K)
 - (II) Ethene
 - (III) n-Hexane (Catalytic aromatization / reforming at 773 K and 10-20 atm)

- Q60.** How will you convert ethanoic acid (acetic acid) into benzene? Write all sequential chemical reactions involved.